

Sparse Transformer Model

Multi-head Attention



Downstream Operators



Unified MHA Module

Row-wise Kernel

- ✓ row-sliced Q parallel
- ✓ sync-elimination
- ✓ shuffle within warp

Block-wise Kernel

- ✓ mask bitmap storage
- ✓ Q register resident
- ✓ async data copying

kernel parameters

Analysis-driven Kernel Selector

Operator Fusion Module

Fusion Scheme Convertor

Hash
Encoding

Numerical
Decoding

Compilation
Template

fusion scheme

numerical expression

Hierarchical Search Engine

Scheme
Initialization

Fusion
Expanding

Parameter
Sampling

Hardware Specification